

1. Create a comparison table listing at least 3 key differences between WAN and LAN, and give one real-world example of each from apps or services you use daily (for example, IRCTC, Instagram, or Zomato).



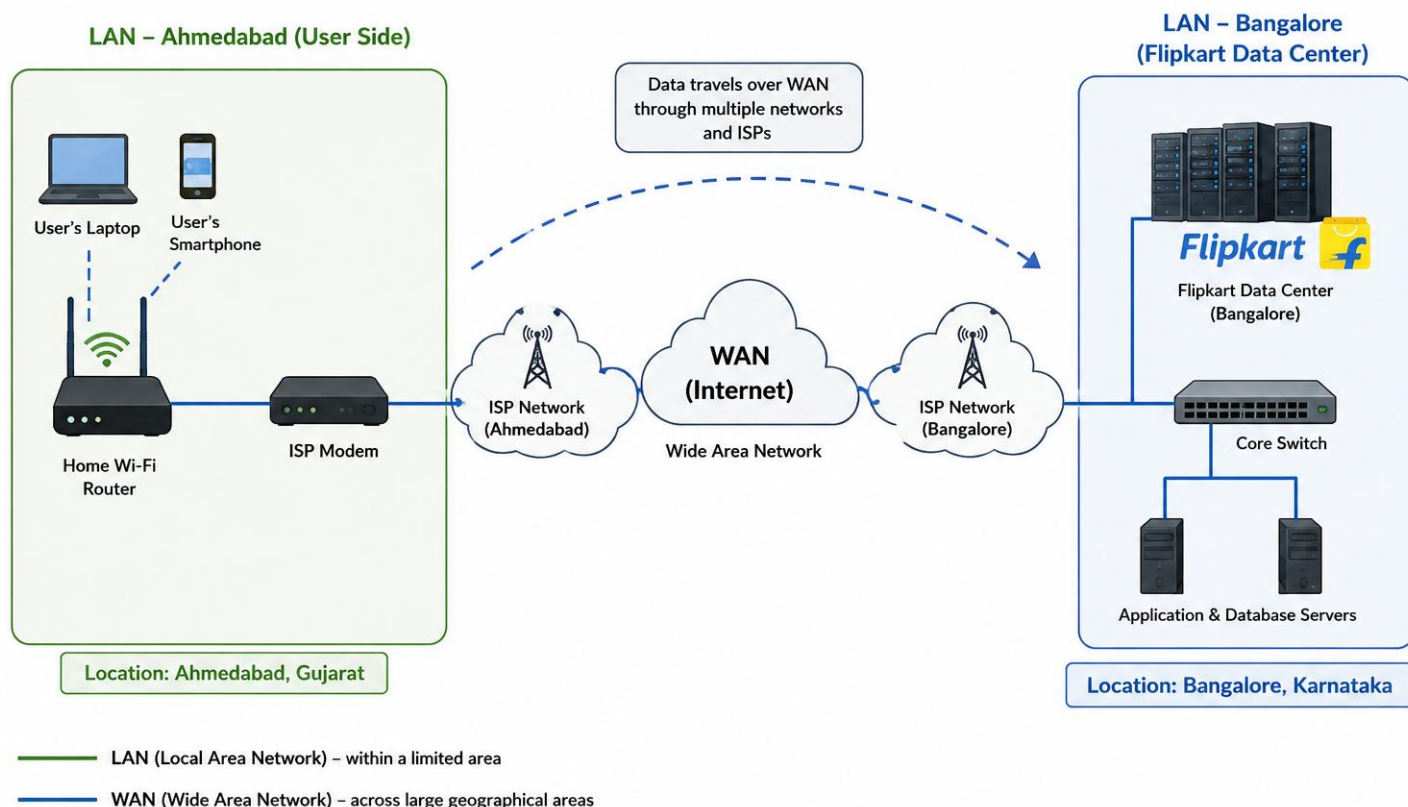
Point	LAN (Local Area Network)	WAN (Wide Area Network)
1. Coverage Area	Covers a small area like a home, office, school, or building.	Covers a large geographical area, such as cities, countries, or the whole world.
2. Speed	Usually faster because devices are connected within a limited area.	Usually slower than LAN because data travels over longer distances and different networks.
3. Ownership	Usually owned and managed by one person or organization.	Often uses infrastructure provided by different organizations or Internet Service Providers.
4. Example	A home Wi-Fi network connecting your laptop and phone to the same router is a LAN.	The Internet is a WAN because it connects networks and users across the world.

Real-World Examples

- **LAN Example:** When you connect your phone and laptop to your home Wi-Fi router and share files between them, you are using a **LAN**.
- **WAN Example:** When you open **Instagram** or **Zomato**, your device communicates with servers located outside your local network through the **Internet (WAN)**.

2. Draw a simple network diagram using any drawing tool (or on paper and upload a photo) that shows how a user in Ahmedabad could connect to a Flipkart data center in Bangalore using a WAN, labeling the LAN and WAN segments clearly.

User in Ahmedabad connecting to Flipkart Data Center in Bangalore using WAN



3. Research and write a short summary (5-7 lines) explaining what MPLS is and how it improves data routing compared to traditional IP routing.

MPLS (**Multiprotocol Label Switching**) is a networking technology that forwards data using short **labels** instead of checking the full IP routing information at every router. In traditional IP routing, each router examines the destination IP address and independently chooses the next hop using its routing table.

With MPLS, packets are assigned to a **Forwarding Equivalence Class (FEC)** and sent through a predefined **Label Switched Path (LSP)**.

Routers can then forward packets by looking at the MPLS label, making forwarding more predictable and efficient.

MPLS also supports **traffic engineering**, allowing network operators to control paths and manage network resources more effectively.

Therefore, MPLS is commonly used in enterprise WANs where reliable, predictable connectivity between different locations is important.

4. **Observe the pre-configured MPLS network in the provided Packet Tracer or GNS3 file, and note down two routing behaviors that are different from standard IP routing.**
Hint: Look for how labels are used and how packets move between routers.



Label Switching (Push, Swap, Pop): Unlike standard IP routing where routers re-evaluate the destination IP at every hop, the ingress router pushes a label onto the packet header. Intermediate routers (Provider/P routers) perform a simple swap of the label, and the egress router pops the label before passing the plain IP packet to its destination LAN.

Pre-Determined Label Switched Paths (LSP): Packets follow a fixed, predetermined path (LSP) through the network core based on label mappings established by signaling protocols like LDP (Label Distribution Protocol), rather than dynamically recalculating the best hop using standard IP routing protocols at every intermediate node.

5. **List three specific benefits MPLS offers over traditional routing for companies running high-traffic apps like YouTube or BookMyShow, and briefly explain how each benefit helps in real-world scenarios.**



1. **Better traffic management** – MPLS can prioritize important data using Quality of Service (QoS). For example, during a big cricket match on **BookMyShow**, ticket requests can be given higher priority so the network stays responsive even when traffic is very high.
2. **Faster and more reliable performance** – MPLS uses predefined paths through the network, which can reduce delays and packet loss compared with ordinary IP routing. For a platform like **YouTube**, this helps videos load smoothly and reduces buffering when millions of users are streaming at the same time.
3. **Easy scalability and reliable connectivity** – MPLS allows companies to connect multiple data centers, offices, and cloud locations through a managed network. As a company grows, new locations or servers can be added without redesigning the entire network. This is useful for services like **BookMyShow** that need their systems to remain available across different regions.